

Supplemental Material for “Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene”

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I. SCOPE OF THE SUPPLEMENTAL MATERIAL

This Supplemental Material documents the executable numerical workflow behind the PRB manuscript. It is organized as a reproducibility and self-audit record: the Bistritzer-MacDonald (BM) model conventions [1], the orbital-projected pairing construction, the finite-band response formula, the figure and table provenance, and the historical PRB benchmark reconstruction are recorded separately.

The main text makes a conservative mechanism claim: an orbital-projected interband-pairing direction produces a reproducible, normalization-conditioned change in the normalized finite-band superfluid response. The present Supplemental Material does not promote the raw response units to final experimental stiffness values. This choice is motivated by the distinction between robust quantum-geometric mechanism diagnostics [2–5] and direct quantitative comparison with recent stiffness experiments [6, 7].

The material below is meant to be read as a set of checks attached to the main text, not as a separate enlargement of the claim. The BM parameters, reporting policy, and filling crosswalk specify what the plotted axes and tabulated labels mean. The pairing-construction and finite-response sections define the diagnostic used in the main text. The provenance section identifies the data and scripts behind each figure and table. The momentum-grid section supplies the selected-grid evidence invoked after the main-text $nk = 9$ table. The benchmark-reconstruction and claim-scope sections record why absolute stiffness, device-level filling, and continuum-limit interpretations are left outside the current mechanism-paper scope.

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TABLE I. Default parameters used for the mechanism scans.

quantity	value
twist angle	$\theta = 1.05^\circ$
Dirac velocity	$v_F = 2.135 \text{ eV \AA}$
AA tunneling	$w_0 = 87.2 \text{ meV}$
AB tunneling	$w_1 = 109.0 \text{ meV}$
reciprocal cutoff	$N_{\text{shell}} = 3$
retained bands	$n_{\text{keep}} = 6$
momentum grid	$nk = 7$ for dense maps, $nk = 9, 11, 13$ for key-point checks
gap scale	$\Delta_0 = 1 \text{ meV}$
velocity finite difference	$dk = 10^{-6} \text{ \AA}^{-1}$

II. BM MODEL AND NUMERICAL PARAMETERS

The normal-state calculation uses the single-valley BM continuum Hamiltonian with orbital basis

$$(tA, tB, bA, bB), \quad (1)$$

where t, b label layers and A, B label sublattices. Unless explicitly stated otherwise, the production mechanism scans use the parameters in Table I.

The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4 \text{ \AA}^2, \quad G_M = 0.054047 \text{ \AA}^{-1}. \quad (2)$$

The raw response conversion used in the audit is

$$D[\text{eV \AA}^2] = \frac{D_{\text{raw}}[\text{meV \AA}^2]}{1000}, \quad (3)$$

with all remaining spin, valley, or convention factors kept explicit rather than absorbed into a hidden prefactor.

A. Reporting Convention for Units and Filling

Table II states the reporting convention used in the present manuscript. The main figures use normalized response ratios, while absolute $\text{eV \AA}^2$ values are restricted to the self-audit

and historical benchmark reconstruction. This is the final observable policy for the current mechanism-paper scope.

TABLE II. Reporting convention for response and filling quantities.

quantity	present use	excluded interpretation
raw response	internal finite-band diagnostic	physical stiffness
$D_{\text{iso}}(\eta)/D_{\text{iso}}(0) - 1$	claim-bearing mechanism observable	absolute stiffness
eV Å ² per flavor	benchmark audit only	experimental comparison
ν_{proxy}	retained-band occupancy label	calibrated carrier density

The retained-band filling proxy is computed as

$$\nu_{\text{proxy}} = g \left[\left\langle N_{\text{keep}}^{-1} \sum_n \Theta(\mu - \varepsilon_{n\mathbf{k}}) \right\rangle_{\mathbf{k}} - \frac{1}{2} \right], \quad (4)$$

with $g = 4$ used as a spin-valley counting factor in the proxy. For the $n_{\text{keep}} = 6$ scans this is not identical to a calibrated experimental filling factor; it is used only to order the chemical-potential scans. The central flat-band crosswalk instead counts the two central BM bands as

$$\nu_{\text{flat}} = g \left[\left\langle \sum_{n \in \text{central pair}} \Theta(\mu - \varepsilon_{n\mathbf{k}}) \right\rangle_{\mathbf{k}} - 1 \right], \quad (5)$$

which ranges from -4 to $+4$ for an empty-to-full central two-band subspace when $g = 4$. On the $nk = 7$, $N_{\text{shell}} = 3$ mesh used for the dense response scan, the central-pair energy window is $[-1.051, 4.134]$ meV. The resulting crosswalk is shown in Table III. It should be read as a flat-band counting reference for the present BM calculation, not as a device-specific carrier-density calibration.

The filling-reference claim is checked by the self-audit

```
python3 scripts/audit_filling_sufficiency.py
```

which writes

```
data/processed/filling_sufficiency_audit.csv
```

Table IV summarizes the result. The crosswalk is sufficient as a BM central-flat-band counting reference for the present mechanism maps. It is not a device-level carrier-density calibration.

TABLE III. Crosswalk between the retained-band filling proxy and central two-flat-band filling for the dense $nk = 7$, $n_{\text{keep}} = 6$ scan.

μ (meV)	ν_{proxy}	ν_{flat}	occupied flat bands/ flavor
-5	-1.483	-4.000	0.000
-4	-1.333	-4.000	0.000
-3	-1.061	-4.000	0.000
-2	-0.857	-4.000	0.000
-1	-0.653	-3.837	0.041
0	-0.381	-2.286	0.429
1	-0.054	-0.327	0.918
2	0.122	0.735	1.184
3	0.395	2.367	1.592
4	0.667	3.918	1.980
5	0.857	4.000	2.000

TABLE IV. Filling-sufficiency self-audit for the current mechanism-paper scope.

check	measured value	manuscript consequence
μ grid	11/11 values	matches dense response map
ν_{proxy}	-1.483 to 0.857	monotone scan label
ν_{flat}	-4.000 to 4.000	full central-pair range
flat-band window	-1.051 to 4.134 meV	bracketed by scan window
sampled fillings	0.327, 0.286, 0.367	near 0, -2 , $+2$ references
Overall scope	8/8 numeric gates	counting reference only

III. ORBITAL-PROJECTED PAIRING CONSTRUCTION

The pairing direction is specified first in the layer-sublattice orbital basis,

$$\Delta_{\text{orb}}(\eta) = \Delta_0 \mathcal{N}_\eta(M_0 + \eta M_1), \quad (6)$$

with

$$M_0 = \tau_0 \sigma_0, \quad M_1 = \tau_x \sigma_x. \quad (7)$$

TABLE V. Raw valley-minus sewing diagnostic. The independently diagonalized valley-minus subspace at $-\mathbf{k}$ is compared with the target $U_+^*(\mathbf{k})$ on an $nk = 3$ grid. Small singular values and large alignment errors indicate that the raw valley-minus eigenvectors should not be used directly to define W_{inter} .

n_{keep}	alignment error	mean minimum singular value	minimum singular value
2	1.3895	0.0267	0.0218
4	1.3782	0.0201	0.0043
6	1.3760	0.0095	0.0016

The two normalization controls used in the manuscript are

$$\text{fixed Frobenius:} \quad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \quad (8)$$

$$\text{fixed } \Delta_0 : \quad \mathcal{N}_\eta = 1. \quad (9)$$

The band-basis pairing matrix is

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_-^*(-\mathbf{k}). \quad (10)$$

The working valley convention in the present diagnostic is the time-reversal sewn proxy

$$U_-(-\mathbf{k}) = U_+^*(\mathbf{k}). \quad (11)$$

The production projection is therefore

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_+(\mathbf{k}). \quad (12)$$

This convention is useful for isolating an orbital-defined interband-pairing signature. It also avoids a known gauge artifact: independently diagonalized valley-minus eigenvectors are not automatically in the same time-reversal sewing convention as the valley-plus eigenvectors. Table V summarizes the raw-basis diagnostic.

This is still a diagnostic convention rather than a full independent two-valley implementation. Valley-asymmetric perturbations, strain, or valley-dependent interactions would require revisiting this sewing choice.

The projected interband-pairing weight is

$$W_{\text{inter}} = \frac{\sum_{n \neq m} |\Delta_{nm}|^2}{\sum_{n,m} |\Delta_{nm}|^2}. \quad (13)$$

For the dense $nk = 7$, $n_{\text{keep}} = 6$ scan, $W_{\text{inter}} \simeq 0.108$ at $\eta = 1$.

TABLE VI. Manuscript artifact provenance. CSV values used in the manuscript tables are checked by `scripts/verify_manuscript_tables.py`.

artifact	data source	script	check
Frobenius heatmap	dense scan CSV	heatmap plotter	figure file exists
fixed- Δ_0 heatmap	dense scan CSV	heatmap plotter	figure file exists
dense response table	eta-summary CSV	eta summarizer	table verifier
$nk = 9$ validation table	key-point summary CSV	eta summarizer	table verifier
PRB audit table	reconstruction CSVs	audit scripts	table verifier

IV. FINITE-BAND RESPONSE AND CHANNEL SPLIT

For a retained band subspace, the finite-band BdG Hamiltonian is

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\text{band}}(\mathbf{k}) \\ \Delta_{\text{band}}^\dagger(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \quad (14)$$

The zero-temperature static current-current response is evaluated as

$$\Lambda_{\alpha\beta} = \sum_{a \neq b} \frac{\langle a | J_\alpha | b \rangle \langle b | J_\beta | a \rangle (f_a - f_b)}{E_b - E_a}. \quad (15)$$

The band-basis velocity is split into diagonal and off-diagonal parts,

$$v_\alpha^{\text{band}} = v_\alpha^{\text{intra}} + v_\alpha^{\text{inter}}. \quad (16)$$

This produces conventional, geometric, and cross-channel response diagnostics. The primary normalized response map uses

$$\delta D_{\text{iso}}(\mu, \eta) = \frac{D_{\text{iso}}(\mu, \eta)}{D_{\text{iso}}(\mu, 0)} - 1, \quad D_{\text{iso}} = \frac{D_{xx} + D_{yy}}{2}. \quad (17)$$

V. FIGURE AND TABLE PROVENANCE

Table VI summarizes the current figure and table provenance. The full machine-readable record is maintained in `notes/figure.table.provenance_2026-06-09.md`.

The dense response data are reproduced by

```
python3 scripts/run_mu_response_scan.py \
  --nk 7 --n-shell 3 --n-keep 6 \
  --mus -5 -4 -3 -2 -1 0 1 2 3 4 5 \
  --etas 0 0.25 0.5 0.75 1 \
  --output data/processed/mu_eta_response_scan_nk7_nkeep6.csv
```

and summarized by

```
python3 scripts/summarize_eta_effect.py \
  data/processed/mu_eta_response_scan_nk7_nkeep6.csv \
  --output data/processed/mu_eta_response_scan_nk7_nkeep6_eta1_summary.csv
```

The table verifier is

```
python3 scripts/verify_manuscript_tables.py
```

VI. KEY-POINT MOMENTUM-GRID CONVERGENCE

The dense maps in the main text use $nk = 7$, while selected key points were checked at $nk = 9$, $nk = 11$, and $nk = 13$. The higher-grid scans use

```
python3 scripts/run_mu_response_scan.py \
  --nk 11 --n-shell 3 --n-keep 6 \
  --mus -4 0 2 4 --etas 0 0.5 1 \
  --output data/processed/mu_response_scan_nk11_nkeep6_keypoints.csv
```

```
python3 scripts/run_mu_response_scan.py \
  --nk 13 --n-shell 3 --n-keep 6 \
  --mus -4 0 2 4 --etas 0 0.5 1 \
  --output data/processed/mu_response_scan_nk13_nkeep6_keypoints.csv
```

and is summarized with `scripts/summarize_eta_effect.py`. The convergence figure is generated with

```
python3 scripts/plot_nk_convergence_keypoints.py
```

This section supports the main-text validation paragraph that follows the $nk = 9$ table. Its role is to test sign and scale stability at selected chemical potentials, not to replace the dense μ - η map or to turn the selected grids into a continuum extrapolation.

Figure 1 and Table VII compare the normalized $\eta = 1$ response relative to $\eta = 0$. The fixed-Frobenius response remains positive at all four key chemical potentials. The fixed- Δ_0 response remains small and sign-sensitive, consistent with the main-text interpretation that the interband-pairing signal is normalization-conditioned.

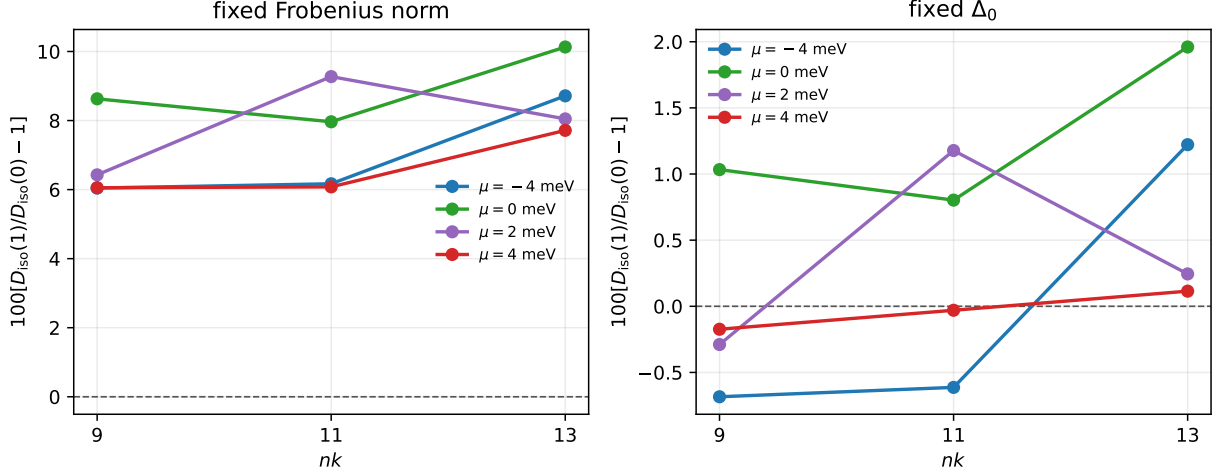


FIG. 1. Selected-grid convergence of the normalized $\eta = 1$ response. The left panel shows the fixed-Frobenius comparison used for the main mechanism claim; the right panel shows the fixed- Δ_0 control. The plotted values are drawn from the same summary data tabulated in Table VII.

TABLE VII. Key-point momentum-grid check for $D_{\text{iso}}(\eta = 1)/D_{\text{iso}}(\eta = 0) - 1$.

mode, μ (meV)	$nk = 9$	$nk = 11$	$nk = 13$	$13 - 9$
fixed Δ_0 , -4	-0.0068	-0.0061	+0.0122	+0.0191
fixed Δ_0 , 0	+0.0103	+0.0080	+0.0196	+0.0093
fixed Δ_0 , 2	-0.0029	+0.0118	+0.0024	+0.0053
fixed Δ_0 , 4	-0.0017	-0.0003	+0.0011	+0.0029
fixed Frobenius, -4	+0.0604	+0.0617	+0.0872	+0.0267
fixed Frobenius, 0	+0.0863	+0.0796	+0.1013	+0.0150
fixed Frobenius, 2	+0.0643	+0.0927	+0.0805	+0.0162
fixed Frobenius, 4	+0.0605	+0.0608	+0.0772	+0.0166

Across the three key-point grids the fixed-Frobenius response remains positive, with the largest $nk = 13 - nk = 9$ shift about 2.67 percentage points. Therefore the current manuscript claims sign and order-of-magnitude stability of the normalized mechanism signal, not a strict continuum-limit extrapolation of the tabulated values.

As an additional finite-grid audit, the same values are fitted to

$$y(nk) = y_0 + a/nk^2, \quad y = 100 [D_{\text{iso}}(\eta = 1)/D_{\text{iso}}(\eta = 0) - 1]. \quad (18)$$

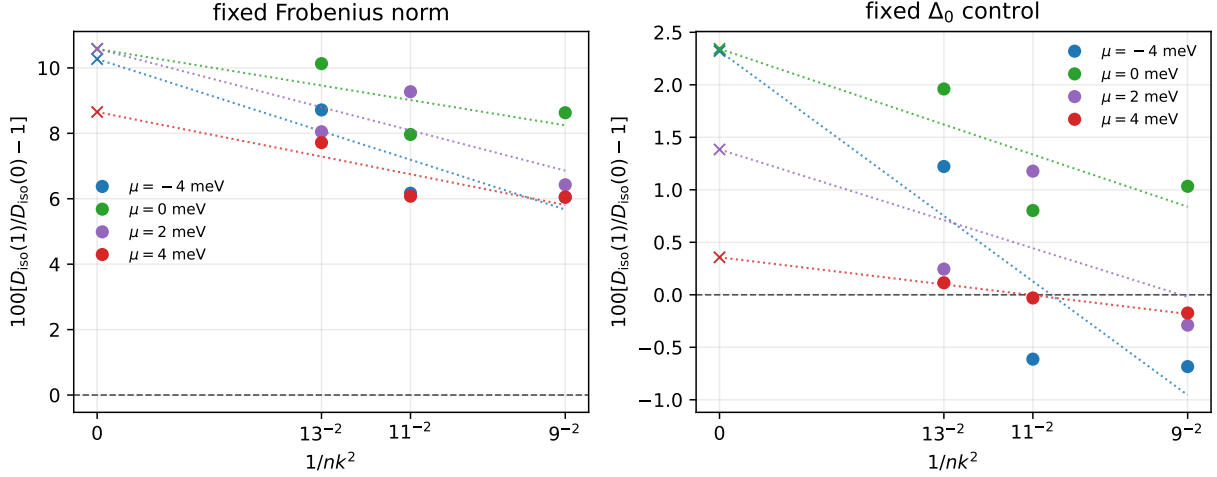


FIG. 2. Finite-grid trend audit for the selected $nk = 9, 11, 13$ key points. Markers at nonzero $1/nk^2$ are computed values, dotted lines are linear $1/nk^2$ fits, and crosses at zero are fitted intercepts. The fixed-Frobenius trend remains positive for all four chemical potentials; the fixed- Δ_0 control remains much smaller and sign-sensitive on the measured grids.

TABLE VIII. Finite-grid trend audit for the selected key points. Values are in percent. The $1/nk^2$ intercept is the plotted trend marker, while the $1/nk$ intercept is a sensitivity check.

mode, μ (meV)	min	max	$1/nk^2$ int.	$1/nk$ int.	sign
fixed Δ_0 , -4	-0.68	1.22	2.32	4.81	mixed
fixed Δ_0 , 0	0.80	1.96	2.34	3.53	positive
fixed Δ_0 , 2	-0.29	1.18	1.38	2.20	mixed
fixed Δ_0 , 4	-0.17	0.11	0.36	0.74	mixed
fixed Frobenius, -4	6.04	8.72	10.27	13.77	positive
fixed Frobenius, 0	7.96	10.13	10.58	12.48	positive
fixed Frobenius, 2	6.43	9.27	10.58	12.94	positive
fixed Frobenius, 4	6.05	7.72	8.65	10.82	positive

Figure 2 shows the measured points, the $1/nk^2$ trend lines, and the intercept markers at $1/nk^2 = 0$. Table VIII also lists a $1/nk$ intercept as a simple sensitivity check. With only three grids, these intercepts are an audit of trend stability rather than a final continuum extrapolation. The fixed-Frobenius intercepts remain positive in both trend coordinates,

TABLE IX. Targeted $nk = 15$ spot check for representative key points.

mode, μ (meV)	$nk = 13$	$nk = 15$	$15 - 13$	$W_{\text{inter}}(15)$
fixed Δ_0 , 0	+0.0196	+0.0085	-0.0111	0.1076
fixed Δ_0 , 2	+0.0024	+0.0071	+0.0046	0.1076
fixed Frobenius, 0	+0.1013	+0.0897	-0.0116	0.1076
fixed Frobenius, 2	+0.0805	+0.0865	+0.0060	0.1076

TABLE X. Convergence-sufficiency self-audit for the current mechanism-paper scope.

check	measured value	manuscript consequence
Frob. sign	6.04 to 10.13%	sign stable
Frob. spread	2.85 pp	scale stable
Trend intercepts	8.65% / 10.82%	positive trend audits
$nk = 15$ spot check	8.97%, 8.65%; shift 1.16 pp	central spot stable
fixed- Δ_0 control	max 1.96%; mixed signs	weak/sign-sensitive
Overall scope	8/8 numeric gates	selected-grid claim only

whereas the fixed- Δ_0 measured-grid signs remain mixed except at $\mu = 0$.

A targeted higher-grid spot check was also run at $nk = 15$ for $\mu = 0$ and 2 meV, using only $\eta = 0$ and $\eta = 1$. This is not a full dense-grid replacement, but it tests whether the two central representative key points remain in the same response range on a denser mesh. Table IX shows that the fixed-Frobenius response remains positive and close to the $nk = 13$ values, while the fixed- Δ_0 control remains below one percent at both chemical potentials.

The selected-grid evidence is finally summarized by the convergence-sufficiency self-audit

```
python3 scripts/audit_convergence_sufficiency.py
```

which writes

```
data/processed/convergence_sufficiency_audit.csv
```

Table X records the manuscript-level consequence. The audit supports the selected-grid normalized mechanism claim used in the main text. It does not constitute a dense-mesh replacement or a strict continuum-limit numerical value.

VII. HISTORICAL PRB BENCHMARK RECONSTRUCTION

The production interband-pairing scan and the historical PRB table reconstruction are kept separate. The best simple unified reconstruction route for the old uniform- s benchmark is

$$D_{\text{conv}} = 2D_{\text{intra},\tau_z}, \quad D_{\text{geom}} = D_{\text{inter},\tau_z}. \quad (19)$$

For $n_{\text{keep}} = 6$, this gives $D_{\text{total}} = 126.66 \text{ eV } \text{\AA}^2$ versus the historical target $129.30 \text{ eV } \text{\AA}^2$, and $D_{\text{geom}} = 75.22 \text{ eV } \text{\AA}^2$ versus $75.30 \text{ eV } \text{\AA}^2$.

The old $n_{\text{keep}} = 2$ endpoint is more convention-sensitive. The closest audited reconstruction combines a Gamma-centered mesh, central-pair band selection, an all- τ_z current convention, a factor of two in the intraband term, and a full curvature-like conventional correction. This gives

$$D_{\text{total}} = 66.18 \text{ eV } \text{\AA}^2 \quad \text{versus} \quad 67.50 \text{ eV } \text{\AA}^2. \quad (20)$$

Because this full-curvature correction does not work uniformly at $n_{\text{keep}} = 6$, the reconstruction is treated as a historical audit rather than a production convention.

VIII. CLAIM SCOPE AND EXCLUDED INTERPRETATIONS

The manuscript is framed as a normalized-response mechanism paper. The text-level claim-scope audit is generated with

```
python3 scripts/audit_claim_scope.py
```

and the observable-policy audit is generated with

```
python3 scripts/audit_observable_policy.py
```

The audit outputs are

```
data/processed/claim_scope_audit.csv
```

```
data/processed/observable_policy_audit.csv
```

They are not numerical validation tools; they check that the manuscript retains the boundary statements needed to avoid overinterpreting the finite-band diagnostic calculation. Table [XI](#) summarizes the current claim scope.

TABLE XI. Allowed manuscript claims and excluded interpretations.

topic	allowed in the present manuscript	excluded interpretation
normalized response mechanism signature		absolute stiffness enhancement
fixed- Δ_0 control	weak/sign-sensitive control	universal enhancement
momentum grids	selected sign/scale stability	continuum-limit value
filling variable	central flat-band crosswalk	device density calibration
absolute units	benchmark self-audit only	experimental comparison
valley partner	tr_sewn diagnostic convention	valley-asymmetric calculation

IX. CURRENT LIMITATIONS AND NEXT VALIDATION GATES

The current submission-critical gaps are:

1. a full continuum-grid extrapolation beyond the present selected $nk = 9$, $nk = 11$, $nk = 13$ trend audit and targeted $nk = 15$ spot check;
2. a fully independent two-valley implementation before studying valley-asymmetric perturbations;
3. a final absolute stiffness convention only before any direct experimental comparison;
4. a device-level filling calibration beyond the central-flat-band crosswalk reported here.

These limitations do not invalidate the normalized-response mechanism claim, but they define the boundary of the present PRB mechanism submission. Direct experimental-comparison claims require further convergence, calibration, and absolute-convention work.

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DECLARATIONS

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Conflict of interest: The author declares no conflict of interest.

Data availability: All data and code supporting this work are available from the corresponding author upon reasonable request.

Ethics approval: Not applicable.

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- [1] R. Bistritzer and A. H. MacDonald, Moiré bands in twisted double-layer graphene, [Proceedings of the National Academy of Sciences](#) **108**, 12233 (2011).
 - [2] S. Peotta and P. Törmä, Superfluidity in topologically nontrivial flat bands, [Nature Communications](#) **6**, 8944 (2015).
 - [3] X. Hu, T. Hyart, D. I. Pikulin, and E. Rossi, Geometric and conventional contribution to the superfluid weight in twisted bilayer graphene, [Physical Review Letters](#) **123**, 237002 (2019).
 - [4] A. Julku, T. J. Peltonen, L. Liang, T. T. Heikkilä, and P. Törmä, Superfluid weight and Berezinskii-Kosterlitz-Thouless transition temperature of twisted bilayer graphene, [Physical Review B](#) **101**, 060505 (2020).
 - [5] F. Xie, Z. Song, B. Lian, and B. A. Bernevig, Topology-bounded superfluid weight in twisted bilayer graphene, [Physical Review Letters](#) **124**, 167002 (2020).
 - [6] H. Tian, X. Gao, Y. Zhang, *et al.*, Evidence for Dirac flat band superconductivity enabled by quantum geometry, [Nature](#) **614**, 440 (2023).
 - [7] M. Tanaka, J. I.-J. Wang, T. H. Dinh, *et al.*, Superfluid stiffness of magic-angle twisted bilayer graphene, [Nature](#) **638**, 99 (2025).